

Online Appendix for “Identifying Demand with Network Externalities”^{*}

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O.1 Verification that G_Δ is a distribution function

Because F is strictly increasing, its quantile function F^{-1} is strictly increasing on $(0, 1)$. For $\Delta \geq 0$, the function $G_\Delta^{-1}(s) = F^{-1}(s) + \Delta s$ is strictly increasing and therefore admits a well-defined and strictly increasing inverse G_Δ . Moreover, $\lim_{s \rightarrow 0} F^{-1}(s) = -\infty$ and $\lim_{s \rightarrow 1} F^{-1}(s) = \infty$. As Δs is bounded on $(0, 1)$, the same limits hold for $G_\Delta^{-1}(s)$, implying $\lim_{x \rightarrow -\infty} G_\Delta(x) = 0$ and $\lim_{x \rightarrow +\infty} G_\Delta(x) = 1$. Right-continuity of G_Δ follows from the continuity of F . Hence G_Δ is a valid cumulative distribution function.

O.2 Proof of Proposition 2

Proof. First, invert equation (7) to obtain

$$\bar{\sigma}_j^{-1}(S_t) = X_{jt} + h_j(S_t) + \Xi_{jt}.$$

This operation is legal by Assumption INVERT-MARKET. Next, rearrange terms to obtain

$$X_{jt} = \underbrace{\bar{\sigma}_j^{-1}(S_t) - h_j(S_t)}_{=\kappa_j(S_t)} - \Xi_{jt}.$$

Let $\kappa_j(S_t) = \bar{\sigma}_j^{-1}(S_t) - h_j(S_t)$. By Assumption IV-MARKET as applied with $V_t = S_t$ and the nonparametric identification argument of Newey and Powell (2003), κ_j is identified on the support of S_t . Thus, each Ξ_{jt} is identified as

$$\Xi_{jt} = \kappa_j(S_t) - X_{jt},$$

so that the vector Ξ_t is identified. Note that X_t and Ξ_t are the only elements that implicitly determine market shares via the fixed point in equation (3). Consequently, with X_t and S_t observed and Ξ_t identified, the mapping from (X_t, Ξ_t) to the observable solutions S_t of (3) is identified. $\square$

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O.3 Proof of Proposition 3

Proof. Assumption INVERT-MARKET implies the existence of an inverse $\bar{\sigma}^{-1}$ of the average choice probability function $\bar{\sigma}$ that satisfies, for each j ,

$$\bar{\sigma}_j^{-1}(S_t) = X_{jt} + h_j(M_t S_t) + \Xi_{jt}.$$

Rearranging terms in the equation above yields a nonparametric regression equation:

$$X_{jt} = \kappa_j(S_t, M_t) - \Xi_{jt}$$

for

$$\kappa_j(s, m) = \bar{\sigma}_j^{-1}(s) - h_j(ms). \quad (1)$$

Assumption IV-MARKET as applied with $V_t = (S_t, M_t)$ and the identification argument of Newey and Powell (2003) yield the identification of κ_j on the support of $\{S_t, M_t\}$. The two assumptions in the statement of the proposition provide two distinct ways to separately identify $\bar{\sigma}^{-1}$ and h_j . Under assumption (a), we can write $h_j(ms) = h_j(ms_j)$ in a slight abuse of notation and then compute

$$\frac{\partial \kappa_j}{\partial m}(s, m) = -h'_j(ms_j)s_j \quad \Rightarrow \quad h'_j(ms_j) = -\frac{1}{s_j} \frac{\partial \kappa_j}{\partial m}(s, m).$$

Given that the right-hand side of the second equation is a function of observed and identified quantities, h'_j is identified on the support of $M_t S_{jt}$. Differentiation is legal here by Assumption CALC-TOT. The entire function h_j is identified under the location normalization provided in the main text that $h_j(q_j^*) = 0$ for all j for a known vector q^* . Indeed, for q in the support of $M_t S_t$,

$$h_j(q_j) = \int_{q_j^*}^{q_j} h'_j(\tilde{q}) d\tilde{q}$$

by the fundamental theorem of calculus and $h(q_j^*) = 0$. Integration of h' over $[q_j^*, q_j]$ is justified by the convexity of the support of $M_t S_t$ as stipulated by Assumption CALC-TOT. The function $\bar{\sigma}_j^{-1}$ is then immediately identified on the support of s_t for each j , which implies the identification of $\bar{\sigma}$ on the support of $\delta(X_t, M_t S_t, \Xi_t)$. This completes the identification argument under assumption (a).

Now consider assumption (b). Note that

$$\frac{\partial \kappa_j}{\partial s}(s, m) = \nabla \bar{\sigma}_j^{-1}(s) - m \nabla h_j(ms), \quad (2)$$

where ∇ is the gradient operator. Differentiation is legal here by Assumption CALC-TOT; the differentiability and invertibility of $\bar{\sigma}$ together imply the differentiability of $\bar{\sigma}^{-1}$. We have

$$\lim_{m \downarrow 0} \frac{\partial \kappa_j}{\partial s}(s, m) = \nabla \bar{\sigma}_j^{-1}(s), \quad (3)$$

which shows how the large support assumption leads to identification of $\nabla \bar{\sigma}_j^{-1}$; the right-hand side of (3) is identified because (i) κ_j is identified on the full support of $\{S_t, M_t\}$ and (ii) the closure of this support includes $M_t = 0$ for all S_t in its support. Note that the identification of ∇h_j consequently follows from (2). The levels of $\bar{\sigma}_j^{-1}$ and h_j are subsequently identified using the location normalization that $h_j(q_j^*) = 0$ for a known vector q_j^* . Indeed, the fundamental theorem of calculus for line integrals implies that for a differentiable path r with $r(0) = q_j^*$ and $r(1) = q_j$ in the support of $M_t S_t$,

$$\int_0^1 \nabla h_j(r(\tau)) \cdot r'(\tau) d\tau = h_j(q).$$

Integration of ∇h_j over a path r in the support of $M_t S_t$ is justified by the convexity of this support as stipulated by Assumption CALC-TOT. Given the definition of κ_j in (1), the identification of h_j on the support of $M_t S_t$ and the previously established identification of κ_j on the support of $\{S_t, M_t\}$ implies the identification of $\bar{\sigma}^{-1}$ on the support of s_t . The identification of $\bar{\sigma}^{-1}$ on the support of s_t straightforwardly implies the identification of $\bar{\sigma}$ on the support of $\delta(X_t, S_t, \Xi_t)$. $\square$

Identification under the relaxed index assumption

$$\begin{aligned} \bar{J}_{jt}(X_t, M_t S_t, \Xi_t) &= \bar{\sigma}_j(\delta(X_t, \Xi_t), M_t S_t) \\ \delta_j(X_t, \Xi_t) &= X_{jt} + \Xi_{jt}. \end{aligned} \tag{4}$$

is generally only possible when there is a single product ($J = 1$). Condition (b) in Proposition 3 no longer achieves identification because the inverse demand function $\bar{\sigma}^{-1}$ depends on $M_t S_t$; sending $M_t \rightarrow 0$ allows the researcher to learn about the values of $\bar{\sigma}^{-1}$ at $M_t = 0$ but not at other values as under the stronger index assumption, which permits extrapolation. The result is formally presented below in Proposition 1, which additionally requires adapted versions of Assumptions INVERT-QUANTITIES and CALC-TOT.

Assumption INVERT-QUANTITIES-2 (Invertible demand — quantities model). For all q in the support of $M_t S_t$, the function $\bar{\sigma}(\cdot, q)$ is injective on the support of $\delta(X_t, \Xi_t)$.

Assumption CALC-TOT-2 (Technical conditions for total quantities model). (i) $\bar{\sigma}$ is continuously differentiable; and (ii) the support of $\{S_t, M_t S_t\}$ is convex.

Part (ii) strengthens the corresponding condition of Assumption CALC-TOT from the support of $M_t S_t$ to the joint support of shares and quantities, because the proof below recovers $\bar{\sigma}^{-1}$ by integrating its partial derivatives with respect to both arguments along a path in that joint support.

It is also necessary to adapt the location normalization. Here, I normalize that $\bar{\sigma}^{-1}(s^0, q^0) = 0$ at a known point (s^0, q^0) in the support of $\{S_t, M_t S_t\}$.

Because quantities no longer enter through the additively separable index, the relaxed model

contains no separate network externality function h_j ; its content is absorbed into the dependence of $\bar{\sigma}$ on its second argument. The proposition below therefore identifies the composite function $\bar{\sigma}$ rather than h and $\bar{\sigma}$ separately, so that the network externality cannot be disentangled from the shape of demand even when $J = 1$.

Proposition 1. *Suppose that Assumptions INVERT-QUANTITIES-2, IV-MARKET, and CALC-TOT-2 hold. Additionally suppose that $J = 1$ (i.e., there is one product). Then $\bar{\sigma}^{-1}$ is identified on the support of $\{S_t, M_t S_t\}$, $\bar{\sigma}$ is identified on the support of $\{\delta(X_t, \Xi_t), M_t S_t\}$, and Ξ_t is identified.*

Proof. By Assumption INVERT-QUANTITIES-2, equation (4) can be inverted to obtain

$$\bar{\sigma}_j^{-1}(S_t, M_t S_t) = X_{jt} + \Xi_{jt}.$$

By the nonparametric instrumental variables approach invoked in the proof of Proposition 3, the function κ_j defined by $\kappa_j(s, m) = \bar{\sigma}_j^{-1}(s, ms)$ is identified on the support of (s, ms) . It remains to recover $\bar{\sigma}_j^{-1}(s, ms)$ from κ_j . Now invoke the proposition's hypothesis that $J = 1$. Then, taking derivatives,

$$\frac{\partial \kappa_j}{\partial m}(s, m) = \frac{\partial \bar{\sigma}_j^{-1}}{\partial (ms)}(s, ms)s.$$

The re-arrangement of terms

$$\frac{\partial \bar{\sigma}_j^{-1}}{\partial (ms)}(s, ms) = \frac{1}{s} \frac{\partial \kappa_j}{\partial m}(s, m),$$

establishes that $\frac{\partial \bar{\sigma}_j^{-1}}{\partial (ms)}(s, ms)$ is identified on the support of (s, ms) , as each object on the right-hand side is either identified or observed. Next, note that

$$\frac{\partial \kappa_j}{\partial s}(s, m) = \frac{\partial \bar{\sigma}_j^{-1}}{\partial s}(s, ms) + \frac{\partial \bar{\sigma}_j^{-1}}{\partial (ms)}(s, ms)m$$

and hence

$$\frac{\partial \bar{\sigma}_j^{-1}}{\partial s}(s, ms) = \frac{\partial \kappa_j}{\partial s}(s, m) - \frac{\partial \bar{\sigma}_j^{-1}}{\partial (ms)}(s, ms)m.$$

This identifies $\frac{\partial \bar{\sigma}_j^{-1}}{\partial s}(s, ms)$ on the support of (s, ms) as the objects on the right-hand side are either observed or identified. The identification of both $\frac{\partial}{\partial s} \bar{\sigma}_j^{-1}(s, ms)$ and $\frac{\partial}{\partial (ms)} \bar{\sigma}_j^{-1}(s, ms)$ identifies $\bar{\sigma}_j^{-1}$ itself. Indeed, by the fundamental theorem of calculus for line integrals, the value of $\bar{\sigma}_j^{-1}$ at any point of the support of $\{S_t, M_t S_t\}$ is recovered by integrating the two partial derivatives along a path from (s^0, q^0) to that point; such a path lies in the support by the convexity stipulated in part (ii) of Assumption CALC-TOT-2, and the location normalization $\bar{\sigma}_j^{-1}(s^0, q^0) = 0$ fixes the level. Finally, $\Xi_{jt} = \bar{\sigma}_j^{-1}(S_t, M_t S_t) - X_{jt}$, so Ξ_t is identified. This completes the identification argument. $\square$

O.4 Proof of Proposition 4

This appendix proves Proposition 4 by adapting the identification argument of Berry and Haile (2024) to the setting in which market shares rather than prices enter choice probabilities as endogenous variables. The structure follows their Lemmas 1–4 and Theorems 1 and 2. Throughout, Assumptions IND-MICRO, INV-DEMAND, INV-INDEX, SUPP, NOND, and TECH are maintained. Assumption IV-MICRO is invoked where stated.

Preliminaries. Following Section 2.5 of Berry and Haile (2024), for each $x \in \text{supp } X_t$ I fix a point $w^0(x) \in \mathcal{W}(x)$ (by Assumption TECH, almost every point of $\mathcal{W}(x)$ is valid for this purpose) and impose the location and scale normalizations $\Gamma(w^0(x), x) = 0$ and $\partial\Gamma(w, x)/\partial w|_{w=w^0(x)} = I$. Because any invertible transformation of the index can be undone by a corresponding transformation of σ , these place no restriction on choice probabilities. The observed market- t conditional choice probability functions are $\varsigma_t(w) = \sigma(\Gamma(w, x_t) + \xi_t, s_t, x_t)$, where (s_t, x_t, ξ_t) denote market t 's realizations of (S_t, X_t, Ξ_t) .

For $(s, x, \xi) \in \text{supp}\{S_t, X_t, \Xi_t\}$, define

$$\mathcal{S}(s, x, \xi) = \sigma(\Gamma(\mathcal{W}(x), x) + \xi, s, x), \quad \mathcal{S}(s, x) = \bigcup_{\xi \in \text{supp } \Xi_t | \{S_t=s, X_t=x\}} \mathcal{S}(s, x, \xi).$$

The set $\mathcal{S}(s, x, \xi)$ combines the consumer-level choice probability vectors arising within a market with realizations (s, x, ξ) as consumer characteristics range over $\mathcal{W}(x)$. The set $\mathcal{S}(s, x)$ collects those arising across all markets with realized $(S_t, X_t) = (s, x)$. By Assumptions INV-DEMAND and INV-INDEX, for each $\varsigma \in \mathcal{S}(s, x, \xi)$, there is exactly one $w \in \mathcal{W}(x)$ with $\sigma(\Gamma(w, x) + \xi, s, x) = \varsigma$. Denote this value by $w^*(\varsigma; s, x, \xi)$ and write $w_t^*(\varsigma) = w^*(\varsigma; s_t, x_t, \xi_t)$.

As in Berry and Haile (2024), two observations drive the proof. First, in each market t , the set $\mathcal{S}(s_t, x_t, \xi_t)$ and the function w_t^* are directly observed even though ξ_t is not. This is because w_t^* is the inverse of the observed function ς_t . It is continuously differentiable almost everywhere by Assumption TECH and the inverse function theorem. Second, applying $\sigma^{-1}(\cdot; s, x)$ (which is well defined by Assumption INV-DEMAND) to the equation defining w^* yields

$$\Gamma(w^*(\varsigma; s, x, \xi), x) + \xi = \sigma^{-1}(\varsigma; s, x) \tag{5}$$

for all $(s, x, \xi) \in \text{supp}\{S_t, X_t, \Xi_t\}$ and $\varsigma \in \mathcal{S}(s, x, \xi)$.

Lemma 1. *For each $x \in \text{supp } X_t$, there exist $s \in \text{supp } S_t \mid X_t = x$ and $\Delta > 0$ such that, for all w and w' in $\mathcal{W}(x)$ satisfying $\|w' - w\| < \Delta$, there exist a choice probability vector ς and vectors ξ and ξ' in $\text{supp } \Xi_t \mid \{S_t = s, X_t = x\}$ such that $w = w^*(\varsigma; s, x, \xi)$ and $w' = w^*(\varsigma; s, x, \xi')$. Furthermore, such (s, Δ) are identified.*

Proof. Fix $x \in \text{supp } X_t$. By part (a) of Assumption NOND, there exists $s \in \text{supp } S_t \mid X_t = x$ such that $\text{supp } \Xi_t \mid \{S_t = s, X_t = x\}$ is open and therefore contains a ball of some radius

$\epsilon > 0$ centered at a point $\bar{\xi}$. Take any w and w' in $\mathcal{W}(x)$ with

$$\|\Gamma(w', x) - \Gamma(w, x)\| < \epsilon, \quad (6)$$

and set $\xi' = \bar{\xi}$ and $\xi = \bar{\xi} + \Gamma(w', x) - \Gamma(w, x)$, both of which lie in $\text{supp } \Xi_t \mid \{S_t = s, X_t = x\}$. By construction, $\Gamma(w, x) + \xi = \Gamma(w', x) + \xi'$, so setting

$$\varsigma = \sigma(\Gamma(w, x) + \xi, s, x) = \sigma(\Gamma(w', x) + \xi', s, x),$$

the definition of w^* yields $w = w^*(\varsigma; s, x, \xi)$ and $w' = w^*(\varsigma; s, x, \xi')$. By the uniform continuity of $\Gamma(\cdot, x)$, there exists $\Delta > 0$ such that (6) holds whenever $\|w' - w\| < \Delta$. Finally, a candidate pair (s, Δ) satisfies the conclusion of the lemma if and only if, for all w and w' in $\mathcal{W}(x)$ with $\|w' - w\| < \Delta$, there exist a choice probability vector ς and markets t and t' with $(S_t, X_t) = (S_{t'}, X_{t'}) = (s, x)$ such that $w = w_t^*(\varsigma)$ and $w' = w_{t'}^*(\varsigma)$. Every object appearing in this condition is observed, so such (s, Δ) are identified. $\square$

Lemma 2. *For every $x \in \text{supp } X_t$, there exists a known $\Delta > 0$ such that, for almost all w and w' in $\mathcal{W}(x)$ satisfying $\|w' - w\| < \Delta$, the matrix $\left[\frac{\partial \Gamma(w', x)}{\partial w}\right]^{-1} \left[\frac{\partial \Gamma(w, x)}{\partial w}\right]$ is identified.*

Proof. Given x , take identified (s, Δ) as in Lemma 1, and select w and w' in $\mathcal{W}(x)$ with $\|w' - w\| < \Delta$. By Lemma 1, there exist a choice probability vector $\hat{\varsigma}$ and markets t and t' with $(S_t, X_t) = (S_{t'}, X_{t'}) = (s, x)$ such that $w = w_t^*(\hat{\varsigma}) \neq w' = w_{t'}^*(\hat{\varsigma})$, revealing that $\xi_t \neq \xi_{t'}$. The two markets share the same inverse function $\sigma^{-1}(\cdot; s, x)$ because the arguments of σ are the index, market shares, and market characteristics only, and the markets have common realized (s, x) . One point worth emphasizing here is that markets with common (s, x) but $\xi_t \neq \xi_{t'}$ generally differ in their consumer characteristic distributions F_t^W , but F_t^W is not an argument of σ . Thus, the difference across markets in F_t^W does not invalidate the cross-market comparisons that I make below in equation (7). Although ξ_t and $\xi_{t'}$ are latent, the market identities t and t' are observed, as are the functions w_t^* and $w_{t'}^*$. Differentiating (5) with respect to the choice probability vector at $\hat{\varsigma}$ within each of the two markets yields

$$\begin{aligned} \frac{\partial \Gamma(w, x)}{\partial w} \frac{\partial w_t^*(\hat{\varsigma})}{\partial \varsigma} &= \frac{\partial \sigma^{-1}(\hat{\varsigma}; s, x)}{\partial \varsigma}, \\ \frac{\partial \Gamma(w', x)}{\partial w} \frac{\partial w_{t'}^*(\hat{\varsigma})}{\partial \varsigma} &= \frac{\partial \sigma^{-1}(\hat{\varsigma}; s, x)}{\partial \varsigma}. \end{aligned} \quad (7)$$

By Assumption TECH, the Jacobian matrices appearing here are nonsingular almost surely, so for almost all such (w, w') ,

$$\left[\frac{\partial \Gamma(w', x)}{\partial w}\right]^{-1} \left[\frac{\partial \Gamma(w, x)}{\partial w}\right] = \frac{\partial w_{t'}^*(\hat{\varsigma})}{\partial \varsigma} \left[\frac{\partial w_t^*(\hat{\varsigma})}{\partial \varsigma}\right]^{-1}.$$

The matrices on the right-hand side are observed. $\square$

Lemma 3. $\Gamma(\cdot, x)$ is identified on $\mathcal{W}(x)$ for all $x \in \text{supp } X_t$.

Proof. The argument is that of Lemma 3 of Berry and Haile (2024). Fix $x \in \text{supp } X_t$ and take the known $\Delta > 0$ from Lemma 2. For $b \in \mathbb{R}^J$ and $\epsilon > 0$, let $\mathcal{B}(b, \epsilon)$ denote the open ball of radius ϵ centered at b . For each vector of integers $\iota \in \mathbb{Z}^J$, define

$$\mathcal{B}_\iota = \mathcal{W}(x) \cap \mathcal{B}\left(w^0(x) + \frac{\iota \Delta}{2J}, \frac{\Delta}{2}\right).$$

By construction, all w and w' in any given $\mathcal{B}_\iota$ satisfy $\|w' - w\| < \Delta$, so by Lemma 2 the value of $[\partial\Gamma(w', x)/\partial w]^{-1}[\partial\Gamma(w, x)/\partial w]$ is known for almost all w and w' in every $\mathcal{B}_\iota$. Because $\bigcup_{\iota \in \mathbb{Z}^J} \mathcal{B}_\iota$ forms an open cover of $\mathcal{W}(x)$, which is open and connected by Assumption SUPP, for any $w \in \mathcal{W}(x)$ there exists a simple chain of sets $\mathcal{B}_\iota$ linking $w^0(x)$ to w (van Mill, 2002, Lemma 1.5.21). Composing the relative Jacobians along this chain shows that $[\partial\Gamma(w, x)/\partial w]^{-1}[\partial\Gamma(w^0(x), x)/\partial w]$ is known for almost every $w \in \mathcal{W}(x)$, and the normalization $\partial\Gamma(w, x)/\partial w|_{w=w^0(x)} = I$ then delivers $\partial\Gamma(w, x)/\partial w$ itself. The result follows from the fundamental theorem of calculus for line integrals, using the continuity of $\partial\Gamma(\cdot, x)/\partial w$, the connectedness of $\mathcal{W}(x)$, and the boundary condition $\Gamma(w^0(x), x) = 0$. $\square$

Corollary 1. (i) For all $(s, x) \in \text{supp}\{S_t, X_t\}$, $\frac{\partial\sigma^{-1}(\cdot; s, x)}{\partial\varsigma}$ is identified on $\mathcal{S}(s, x)$. (ii) For almost all $(s, x) \in \text{supp}\{S_t, X_t\}$ and all $\varsigma^0, \varsigma^1 \in \mathcal{S}(s, x)$,

$$\Omega(s, x, \varsigma^1, \varsigma^0) \equiv \sigma^{-1}(\varsigma^1; s, x) - \sigma^{-1}(\varsigma^0; s, x)$$

is identified.

Proof. (i) Take any $(s, x) \in \text{supp}\{S_t, X_t\}$ and any market t with $(S_t, X_t) = (s, x)$. Differentiating (5) with respect to ς gives, for all $\varsigma \in \mathcal{S}(s, x, \xi_t)$,

$$\frac{\partial\sigma^{-1}(\varsigma; s, x)}{\partial\varsigma} = \frac{\partial\Gamma(w_t^*(\varsigma), x)}{\partial w} \frac{\partial w_t^*(\varsigma)}{\partial\varsigma}.$$

The first factor on the right-hand side is known by Lemma 3 and the second is observed. Ranging over all markets with realized $(S_t, X_t) = (s, x)$ covers $\mathcal{S}(s, x)$.

(ii) By the fundamental theorem of calculus for line integrals, applied componentwise, part (ii) follows from part (i) provided that any two points of $\mathcal{S}(s, x)$ can be joined by a piecewise smooth path lying within it. An open connected subset of $\mathbb{R}^J$ has this property, any two of its points being joined by a piecewise linear path. I therefore establish that $\mathcal{S}(s, x)$ is open and, for almost all (s, x) , connected.

For openness, fix $\xi \in \text{supp } \Xi_t \mid \{S_t = s, X_t = x\}$ and consider the map $w \mapsto \sigma(\Gamma(w, x) + \xi, s, x)$, whose image is $\mathcal{S}(s, x, \xi)$. It is continuous by Assumption TECH and injective by Assumptions INV-DEMAND and INV-INDEX, as already noted in the preliminaries. Its domain $\mathcal{W}(x)$ is an open subset of $\mathbb{R}^J$ by Assumption SUPP, and W_{it} is $\mathbb{R}^J$ -valued, so the map is a continuous injection from an open subset of $\mathbb{R}^J$ into $\mathbb{R}^J$. Brouwer's invariance of domain therefore implies that $\mathcal{S}(s, x, \xi)$ is open, and $\mathcal{S}(s, x)$ is open as a union of such sets

over ξ .

For connectedness, observe that $\mathcal{S}(s, x)$ is the image of $\mathcal{W}(x) \times (\text{supp } \Xi_t \mid \{S_t = s, X_t = x\})$ under the continuous map $(w, \xi) \mapsto \sigma(\Gamma(w, x) + \xi, s, x)$. The set $\mathcal{W}(x)$ is connected by Assumption SUPP, and $\text{supp } \Xi_t \mid \{S_t = s, X_t = x\}$ is connected for almost all (s, x) by part (b) of Assumption NOND; the product of connected sets is connected, and the continuous image of a connected set is connected. The set $\mathcal{S}(s, x)$ is therefore open and, for almost all (s, x) , connected, which completes the argument. $\square$

Lemma 4. *Let Assumption IV-MICRO hold in addition. Then ξ_{jt} is identified for all j and t .*

Proof. For each $(s, x) \in \text{supp}\{S_t, X_t\}$, let $\varsigma^0(s, x)$ be an arbitrary point in $\mathcal{S}(s, x)$, and for each market t let $\tilde{w}_t$ be an arbitrary point in $\mathcal{W}(x_t)$. Evaluating (5) in market t at $\varsigma = \varsigma_t(\tilde{w}_t)$ gives, for each $j = 1, \dots, J$,

$$\Gamma_j(\tilde{w}_t, x_t) = \sigma_j^{-1}(\varsigma_t(\tilde{w}_t); s_t, x_t) - \xi_{jt}.$$

Using Corollary 1(ii) to rewrite $\sigma_j^{-1}(\varsigma_t(\tilde{w}_t); s_t, x_t)$, this equation becomes

$$\mathcal{Y}_{jt} = \varphi_j(x_t, s_t) - \xi_{jt}, \tag{8}$$

where

$$\begin{aligned} \mathcal{Y}_{jt} &\equiv \Gamma_j(\tilde{w}_t, x_t) - \Omega_j(s_t, x_t, \varsigma_t(\tilde{w}_t), \varsigma^0(s_t, x_t)), \\ \varphi_j(x, s) &\equiv \sigma_j^{-1}(\varsigma^0(s, x); s, x). \end{aligned}$$

By Lemma 3 and Corollary 1, each $\mathcal{Y}_{jt}$ is known. Equation (8) thus takes the form of a separable nonparametric regression model with regressors (X_t, S_t) , unknown regression function φ_j , and structural errors satisfying $\mathbb{E}[\Xi_{jt} \mid X_t, Z_t] = 0$ almost surely by part (i) of Assumption IV-MICRO. Under the completeness condition in part (ii) of Assumption IV-MICRO, identification of φ_j follows from Proposition 2.1 of Newey and Powell (2003), and each $\xi_{jt} = \varphi_j(x_t, s_t) - \mathcal{Y}_{jt}$ is identified in turn. Note also that the arbitrary choice of $\varsigma^0(s, x)$ does not generate spurious overidentification, as a different $\varsigma^0(s, x)$ adds the same function of (x, s) to both $\mathcal{Y}_{jt}$ and φ_j . $\square$

Proof of Proposition 4. By Lemma 4, ξ_t is identified for every market t . Recalling that $\mathcal{J}(C_{it}) = \sigma(\gamma(W_{it}, X_t, \Xi_t), S_t, X_t)$ with $C_{it} = \{W_{it}, S_t, X_t, \Xi_t\}$, and that (W_{it}, S_t, X_t) are observed and all realizations ξ_t are now known, $\mathcal{J}$ is identified on $\text{supp } C_{it}$. $\square$

Remark 1 (Equilibrium multiplicity and selection). The fixed point (3) may admit multiple solutions, in which case the data reflect an unmodelled selection among equilibria,

possibly dependent on Ξ_t . Selection interacts with the identification argument in a limited and interpretable way. First, Lemmas 1–3 and Corollary 1 condition throughout on realized market-level outcomes. Within a market, selection has already resolved, so no step of these arguments involves the selection mechanism. Second, the exclusion restriction in part (i) of Assumption IV-MICRO restricts the joint distribution of the primitives (Ξ_t, X_t, Z_t) . The selection mechanism determines which equilibrium is realized given the primitives; it does not alter the distribution of Ξ_t given (X_t, Z_t) . The exclusion restriction is therefore unaffected by Ξ_t -dependent selection. Third, the completeness condition in part (ii) of Assumption IV-MICRO concerns the joint distribution of (X_t, S_t, Z_t) , an equilibrium object that the selection mechanism helps determine. Completeness is thus a joint restriction on primitives, instruments, and selection. Its role in the proof is nevertheless standard, as the nonparametric instrumental variables step in Lemma 4 does not require knowledge of the source of the endogeneity of S_t . Last, selection bears on the support conditions. Both parts of Assumption NOND restrict $\text{supp } \Xi_t \mid \{S_t, X_t\}$, which reflects the interaction of cross-market variation in F_t^W with equilibrium selection, as discussed in Section 3.

O.5 Identification of models with multiple demographic groups

I now consider the identification of models featuring demographic-group-specific market shares in indirect utilities. Given that the identification of demographic-group-specific market shares typically requires micro data, I limit my attention here to the case in which micro data are available.

Let N_D be the number of demographic groups, which are assumed to partition the set of consumers. I assume here that each demographic group $d \in \{1, \dots, N_D\}$ has distinct product/market unobservables Ξ_{jt}^d , and let $\Xi_t = \{\Xi_{jt}^d\}_{j \in \mathcal{J}, d=1, \dots, N_D}$ denote the full collection of group-specific product-market demand shocks. I denote the market share of product j among consumers of demographic group d in market t by S_{jt}^d and the measure of consumers belonging to demographic group d in market t by M_t^d . Let $S_t = \{S_{jt}^d\}_{j \in \mathcal{J}, d=1, \dots, N_D}$ denote the full collection of demographic-group-specific market shares in market t . The consumer characteristics W_{it} are assumed to vary within demographic groups. Furthermore, let $d(i)$ denote the demographic group of consumer i .

Identification analysis in this setting proceeds similarly as the analysis of micro data setting without distinct demographic groups in the main text. As in the main text, I focus on the shares model, but a similar identification argument applies to the quantities model.

Under suitably generalized assumptions, Proposition 4 is generalized to identify the model with multiple demographic groups. These assumptions are provided below.

Assumption IND-MICRO-DEMO (Index assumption for micro data case). Consumer i 's probability of choosing alternative j conditional on $C_{it} = \{W_{it}, S_t, X_t, \Xi_t, d(i)\}$ is given by

$$\mathcal{J}_j(C_{it}) = \sigma_{j, d(i)}(\gamma^{d(i)}(W_{it}, X_t, \Xi_t^{d(i)}), S_t, X_t)$$

where, for each $d \in \{1, \dots, N_D\}$, $\gamma^d(W_{it}, X_t, \Xi_t^d) = (\gamma_1^d(W_{it}, X_t, \Xi_t^d), \dots, \gamma_J^d(W_{it}, X_t, \Xi_t^d))$ is $\mathbb{R}^J$ -valued with

$$\gamma_j^d(W_{it}, X_t, \Xi_t^d) = \Gamma_j^d(W_{it}, X_t) + \Xi_{jt}^d, \quad \forall j, d.$$

Assumption INV-DEMAND-DEMO (Injectivity of demand). For each d , the $\mathbb{R}^J$ -valued function $\sigma_d(\cdot, s, x) = (\sigma_{1,d}(\cdot, s, x), \dots, \sigma_{J,d}(\cdot, s, x))$ is injective on the support of $\gamma^d(W_{it}, x, \Xi_t^d)$ for all $(s, x) \in \text{supp}\{S_t, X_t\}$.

Assumption INV-INDEX-DEMO (Injectivity of index). For each d , the function $\gamma^d(\cdot, x, \xi)$ is injective on $\text{supp } W_{it} \mid X_t = x, d(i) = d$ for all $(x, \xi) \in \text{supp}\{X_t, \Xi_t^d\}$.

Assumption SUPP-DEMO (Support). For all $x \in \text{supp } X_t$ and each d , the support of W_{it} conditional on $X_t = x$ and $d(i) = d$ is the same in every market and is open and connected.

Assumption NOND-DEMO (Nondegeneracy). (a) For each $x \in \text{supp } X_t$, there exists $s \in \text{supp } S_t \mid X_t = x$ such that, for each d , $\text{supp } \Xi_t^d \mid S_t = s, X_t = x$ is open and connected. (b) For almost every $(s, x) \in \text{supp}\{S_t, X_t\}$ and each d , the set $\text{supp } \Xi_t^d \mid S_t = s, X_t = x$ is connected.

Assumption TECH-DEMO (Technical conditions). For each d and for all $(s, x) \in \text{supp}\{S_t, X_t\}$, (i) $\Gamma^d(\cdot, x)$ is uniformly continuous and continuously differentiable; (ii) $\sigma_d(\cdot, s, x)$ is continuously differentiable; (iii) $\partial \Gamma^d(w, x) / \partial w$ is nonsingular almost surely on $\text{supp } W_{it} \mid X_t = x, d(i) = d$; and (iv) $\partial \sigma_d(\gamma, s, x) / \partial \gamma^d$ is nonsingular almost surely on $\text{supp } \gamma^d(W_{it}, X_t, \Xi_t^d) \mid S_t = s, X_t = x, d(i) = d$.

Assumption IV-MICRO-DEMO (Instruments for micro data case). There exist instrumental variables Z_t satisfying the following assumptions: (i) $\mathbb{E}[\Xi_{jt}^d \mid X_t, Z_t] = 0$ almost surely for all j and d (exclusion restriction); and (ii) for all functions $\Psi(X_t, S_t)$ with finite expectation, $\mathbb{E}[\Psi(X_t, S_t) \mid X_t, Z_t] = 0$ almost surely implies $\Psi(X_t, S_t) = 0$ almost surely (completeness).

The statement of the generalized proposition is as follows.

Proposition 2. *Under Assumptions IND-MICRO-DEMO, INV-DEMAND-DEMO, INV-INDEX-DEMO, SUPP-DEMO, NOND-DEMO, TECH-DEMO, and IV-MICRO-DEMO, the realizations ξ_t^d are identified for all t and d , and the choice probability function s is identified on $\text{supp } C_{it}$.*

Proof. The argument of the proof of Proposition 4 above applies to each demographic group d separately, with the group-specific index γ^d , choice probability function σ_d , conditional choice probability functions, and demand shocks Ξ_t^d in place of their aggregate counterparts, and with the full vector of group-specific market shares S_t —of dimension $J \times N_D$ —as the endogenous conditioning variable throughout. Three adjustments deserve note. First, the

support and dimension conditions on W_{it} (Assumptions INV-INDEX-DEMO and SUPP-DEMO) must hold within each group, so that the analogues of Lemmas 1–3 apply group by group. Second, the completeness condition in Assumption IV-MICRO-DEMO applies to functions of (X_t, S_t) with the enlarged share vector, which generally requires additional instruments, as discussed below. Third, part (b) of Assumption NOND-DEMO must hold for each group's demand shocks conditional on the full share vector, so that the group- d analogue of Corollary 1(ii) applies. $\square$

Note here that the market shares S_t have dimension $J \times N_D$, greater than the J in the case without demographic groups. Thus, additional instrumental variables are generally required; the obvious choice is to use Waldfogel instruments based on distributions of consumer characteristics within market t /demographic group d pairs.

O.6 Microfoundation for network externalities on dating websites

This appendix provides a model of search and matching on a dating website that justifies the inclusion of a network externality term in consumer payoffs from using dating websites. This model is a variant of that proposed by Smith (2006); the model features a mass L of users of a dating website, each of whom finds matches on the site at an exponential rate ρ . Matches are exogenously destroyed at rate δ . Each user's payoffs are discounted at the interest rate r . The flow value to agent i of a match with agent j is f_{ij} and the flow value of remaining unmatched is zero. I assume that f_{ij} are identically and independently distributed across ordered pairs (i, j) from the distribution F . Let $\vartheta = \Pr(f_{ij} > 0)$ and let $\mu_f = \mathbb{E}[f_{ij} | f_{ij} > 0]$. A match forms only by mutual consent, that is, only when f_{ij} and f_{ji} are both positive, which occurs with probability ϑ^2 . Because the two draws are independent, conditioning on j 's consent leaves the distribution of f_{ij} unchanged, so μ_f remains the expected flow value of a match that forms. Suppose that consumer i immediately forms any match that yields a positive flow value f_{ij} and does not account for the option value of waiting for a better match. The average present value¹ of being unmatched is

$$V_u = \frac{\rho \vartheta^2 u}{r} \int [V_m(f) - V_u] dF(f | f > 0), \quad (9)$$

where $u \leq L$ is the mass of unmatched users and $V_m(f)$ is the average present value of being matched with match value f :

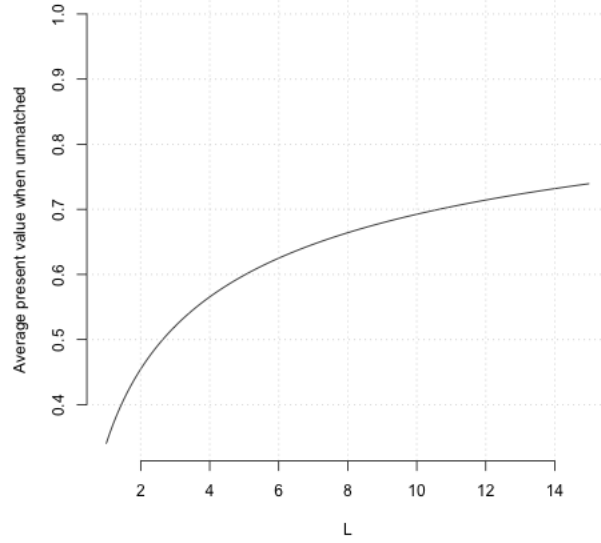
$$V_m(f) = f + \frac{\delta}{r} [V_u - V_m(f)]. \quad (10)$$

We can solve (10) to obtain

$$V_m(f) = \frac{rf + \delta V_u}{r + \delta}$$

¹The average present value is defined as the net present value times the interest rate r .

Figure O.1: Relationship between platform usage and value



and then solve (9) to obtain

$$V_u = \frac{\vartheta^2 u \mu_f}{\psi + \vartheta^2 u}, \quad (11)$$

where $\psi = (r + \delta)/\rho$ is a measure of the search frictions in the market. The steady-state value of u is given by the condition

$$\delta(L - u) = \rho \vartheta^2 u^2,$$

which equates inflows to and outflows from the population of unmatched users. Solving for u yields

$$u = \sqrt{\frac{\delta^2}{4(\vartheta^2 \rho)^2} + \frac{\delta}{\vartheta^2 \rho} L} - \frac{\delta}{2\vartheta^2 \rho}.$$

We can substitute this expression for u into (11) to obtain an expression for the value of being unmatched as a function of the mass of users L on the platform and of other model primitives; this expression is increasing in L . Figure O.1 plots the relationship between V_u and L under the choice of parameters $\rho = 0.5$, $\delta = 0.5$, $\mu_f = 1$, $\vartheta = 1$, and $r = 0.10$. The positive relationship between L and the consumer's value of joining the platform as an unmatched user justifies the inclusion of a network externality term in the indirect utilities of the model considered in the main text.

A two-sided variant. The model above treats the user base as undifferentiated. Consider instead a variant with two sides—labelled men and women—in which matches form only across sides; the same algebra applies to any two-sided partition of users. Masses L_m and L_w of men and women use the site, with unmatched masses u_m and u_w . An unmatched

man now encounters unmatched women at rate ρu_w , so that his match arrival rate is $\rho \vartheta^2 u_w$, and symmetrically for women. The value functions are as above with the opposite side's unmatched mass in place of u :

$$V_u^g = \frac{\vartheta^2 u_{-g} \mu_f}{\psi + \vartheta^2 u_{-g}}, \quad g \in \{m, w\},$$

which is increasing in the mass of unmatched users on the *other* side. Because matches form and dissolve in man–woman pairs, the steady-state conditions

$$\delta(L_m - u_m) = \rho \vartheta^2 u_m u_w = \delta(L_w - u_w)$$

imply $u_m - u_w = L_m - L_w =: \Delta$, and the pair of conditions reduces to a single quadratic in u_w ,

$$\rho \vartheta^2 u_w^2 + (\rho \vartheta^2 \Delta + \delta) u_w - \delta L_w = 0,$$

whose unique nonnegative root is

$$u_w = \sqrt{\left(\frac{\Delta}{2} + \frac{\delta}{2\vartheta^2\rho}\right)^2 + \frac{\delta L_w}{\vartheta^2\rho}} - \left(\frac{\Delta}{2} + \frac{\delta}{2\vartheta^2\rho}\right).$$

Now fix the site's gender composition: let $L_w = \varphi L$ and $L_m = (1 - \varphi)L$ for a total size L and a share $\varphi \in (0, 1)$ of women. Implicitly differentiating the quadratic and substituting the steady-state condition $\rho \vartheta^2 u_w u_m = \delta(\varphi L - u_w)$ yields

$$\frac{\partial u_w}{\partial L} = \frac{\delta(1 - \varphi) u_w}{u_m [\rho \vartheta^2 (u_m + u_w) + \delta]} > 0,$$

and symmetrically $\partial u_m / \partial L > 0$. Holding composition fixed, an increase in total platform size therefore raises the mass of unmatched users on both sides, and so raises both genders' values of using the site.

The implication for the main text is the following: whenever the gender composition of a site's user base is stable across the markets and counterfactuals under study, total usership is a sufficient statistic for the network value of the site—for both genders—with composition absorbed into the constants of the network externality function. The size-based externality f_j of the main text is thus the reduced form of a two-sided matching model in which gender composition matters but does not vary. Two limits of this rationale bear noting. First, the scale effect derives from the increasing returns of the quadratic meeting technology, as in Smith (2006): an unmatched user's meeting rate rises with the opposite side's unmatched mass, whereas under a constant-returns meeting technology, proportional growth of both sides would leave individual match rates unchanged and no size externality would arise. The positive estimated network externality of the main text is consistent with increasing returns in meetings. Second, the argument covers changes in size at stable composition; counterfactuals that alter a site's gender composition would instead require the composition-

Table O.1: Subscription prices and purchase counts by bundle length

Length (months)	eharmony.com		match.com	
	Price	Purchases	Price	Purchases
1	\$56.77	425	\$30.09	434
3	\$76.28	395	\$56.90	167
6	\$159.81	57	\$105.22	98
12	\$169.17	84	—	0

Note: this table reports the mean transaction price for subscriptions of each length and the number of such purchases in the Comscore transaction records for 2007–2008. Subscription lengths are inferred from product names; for purchases whose product name does not reveal the length, I assign the length whose average price is nearest the transaction price. No twelve-month purchases of `match.com` subscriptions appear in the data.

dependent model of Section 3 of the paper, which the data, lacking a gender variable, cannot support.

O.7 Subscription prices

Table O.1 reports, for each paid dating website, the mean price paid for a subscription of each offered length together with the number of purchases of that length appearing in the Comscore transaction records for 2007–2008. The price per month of service declines with subscription length at both sites. These purchases underlie the construction of the price variable P_j described in the main text: per-month prices are averaged across bundle lengths with weights equal to the number of months of service, and the average is expressed as the price of a three-month subscription period, three months being the median subscription length when each subscription is weighted by the number of months that it covers.

O.8 Consumer characteristics

Table O.2 reports summary statistics describing the consumers in the estimation sample constructed in Section 4 of the main text alongside the full panel of Comscore consumers. Consumers in the estimation sample are demographically similar to the Comscore panel as a whole, but they browse the web much more intensively: the median consumer in the estimation sample views roughly twice as many webpages over the year as the median panelist. Users of the four dating websites resemble one another in age and race, although `okcupid.com`’s users are especially intensive internet users and `pof.com`’s users have relatively low incomes.

O.9 Multi-homing

Consumers in my setting are able to use multiple websites, although few consumers multi-home in practice. Among all panelists who use at least one site according to the criterion established by the preceding section, 81% use only one website, 17% use two, and 2% use over two sites. Table O.3 reports shares of each site’s users who use other dating websites. Not only do few panelists multi-home, panelists who multi-home generally spend a large

Table O.2: Summary statistics: consumer characteristics

	All	Est. sample	match	eharmony	pof	okcupid	Outside
Number of consumers	147867	23697	4715	2221	963	281	15517
Age: under 25	2.4	2.5	2.2	2.4	2.5	3.6	2.5
Age: 25–29	5.1	5.1	5.5	5.4	6.3	5.7	4.9
Age: 30–34	9.0	8.6	9.3	8.4	11.2	6.4	8.4
Age: 35–39	10.9	11.5	11.2	11.0	12.5	9.3	11.7
Age: 40–49	31.4	32.2	31.5	32.6	26.9	37.4	32.6
Age: 50 and over	41.2	40.1	40.3	40.3	40.6	37.7	39.9
Income: under \$25k	21.0	19.3	18.9	18.5	20.7	16.7	19.6
Income: \$25k–\$75k	46.0	44.3	44.1	45.6	49.1	42.0	43.9
Income: \$75k–\$100k	14.5	16.0	16.1	16.6	15.4	15.3	16.0
Income: over \$100k	18.5	20.3	20.8	19.4	14.8	26.0	20.6
Education: high school or less	5.1	5.1	5.0	4.3	3.6	7.5	5.3
Education: some college	5.5	5.5	5.9	4.9	5.2	6.8	5.4
Education: college degree	4.4	4.5	4.5	3.6	2.5	5.0	4.7
Education: graduate degree	2.8	3.0	3.3	2.4	1.8	1.1	3.2
Education: unknown	82.2	81.9	81.3	84.8	86.9	79.7	81.4
Race: white	90.9	89.0	92.0	89.3	91.9	90.4	87.8
Race: black	6.3	7.5	5.0	8.1	5.6	3.9	8.4
Race: Asian	1.2	1.5	1.5	1.1	0.4	2.8	1.6
Race: other	1.5	2.0	1.6	1.5	2.1	2.8	2.2
Hispanic	24.6	21.4	21.9	19.5	17.4	16.7	21.8
Children in household	67.9	67.4	66.5	66.8	65.2	67.3	67.9
Broadband connection	89.7	95.9	96.4	94.2	92.9	97.2	96.2
Median annual pages viewed	16608	33257	37140	32089	33059	56732	32105
Median annual browsing hours	305	630	712	593	636	1093	605

Notes: this table describes consumers in the 2007–2008 Comscore data. The “All” column describes all Comscore panelists with valid location information; the “Est. sample” column describes the estimation sample constructed in Section 4 of the main text; and the remaining columns describe the estimation-sample consumers choosing each alternative—i.e., the users of each dating website and the consumers choosing the outside option. Each entry other than those in the first row and the final two rows provides the percentage of the column group’s consumers with the row’s characteristic. Age and education refer to the head of the panelist’s household as recorded by Comscore. The final two rows report medians of panelists’ annual web usage aggregated across all websites.

Table O.3: Multi-homing patterns

Website	Share of panelists using	Share also using			
		eharmony.com	match.com	okcupid.com	pof.com
eharmony	0.35	1.00	0.38	0.02	0.09
match	0.67	0.20	1.00	0.02	0.07
okcupid	0.04	0.19	0.26	1.00	0.19
pof	0.15	0.21	0.32	0.05	1.00

share of their time on a single dating website: the average number of minutes that a multi-homing panelist spends on the panelist's primary site is 1023, whereas the average number of minutes spent on other sites is only 158. In addition, the average share of time that a panelist spends on the panelist's primary site among all dating websites that the panelist uses is 79%; the median across panelists is 81%. Although multi-homing can play an important role in platform competition, the fact that panelists in my sample generally concentrate their online dating activity on a single site motivates my decision to use a model in which each consumer selects a single primary site.

O.10 Robustness: multi-homing

Under the main specification, a consumer uses at most one dating website. This appendix describes a robustness specification in which consumers may use up to two sites. Each set of at most two sites, including the empty set (the outside option), is an alternative. Following the bundle-choice demand structure of Gentzkow (2007), I model the utility of a two-site bundle as the sum of its component sites' utilities (i.e., the site-market effects δ_{jt} and the characteristic effects $W'_{it}\lambda_j$) plus a common two-site term ω , which is positive when sites are complements in demand and negative when they are substitutes. The network externality for a collection of sites A depends on the share of consumers with whom a member of each site in A can interact, namely the users of any site in A , whether single- or multi-homing:

$$f_A(S_t; \eta) = \eta \log S_{At}^{\cup}, \quad S_{At}^{\cup} = \sum_{B: B \cap A \neq \emptyset} S_{Bt}, \quad (12)$$

where S_{Bt} is the share of consumers in market t who use exactly the set B of sites.

In the micro step, I estimate site-market effects δ_{jt} that include the network terms $\eta \log S_{\{j\}t}^{\cup}$ for individual sites j , just as in the main specification. A complication arises for sets A that include two sites, because the sum of the two sites' network terms does not equal the network term for the set $A = \{j, j'\}$:

$$\eta \log S_{\{j\}t}^{\cup} + \eta \log S_{\{j'\}t}^{\cup} \neq \eta \log S_{\{j, j'\}t}^{\cup},$$

both because the network externality function is nonlinear and because $S_{\{j\}t}^{\cup} + S_{\{j'\}t}^{\cup} \neq S_{\{j, j'\}t}^{\cup}$. To address this complication, I decompose the network term for a set of sites $\{j, j'\}$

as follows:

$$\eta \log S_{\{j,j'\}t}^{\cup} = \eta \log S_{\{j\}t}^{\cup} + \eta \log S_{\{j'\}t}^{\cup} + \eta r_{\{j,j'\}t}, \quad r_{\{j,j'\}t} = \log S_{\{j,j'\}t}^{\cup} - \log S_{\{j\}t}^{\cup} - \log S_{\{j'\}t}^{\cup},$$

where $r_{\{j,j'\}t} \geq 0$ is computed from observed shares. Accordingly, I specify the utility of the pair $\{j, j'\}$ as $\delta_{jt} + \delta_{j't} + W'_{it}(\lambda_j + \lambda_{j'}) + \omega + \eta r_{\{j,j'\}t}$. Given η , the last term is a known constant in the micro-step likelihood, so the micro step estimates $(\delta, \lambda, \omega)$ exactly as before. Estimation alternates between the two steps: starting from the main specification's estimate of η , I estimate the micro step given η , re-estimate η in the market step from the resulting δ_{jt} , and repeat until successive estimates of η differ by less than 10^{-4} .

To operationalize this model, I treat consumers who satisfy the main text's usage criteria for two sites as users of both. For consumers who satisfy the criteria for more than two sites, I take the two to be the consumer's primary site, defined as in the main text, and the remaining qualifying site on which the consumer viewed the most pages. Consumers who satisfy the criteria for one site or none are assigned as in the main specification.

I estimate the micro step of the specification by maximum likelihood and find $\hat{\omega} = -1.42$ (standard error 0.04), indicating that sites are substitutes in preferences. The market step estimates appear in Table O.4: the network externality parameter estimate is $\hat{\eta} = 0.73$ (standard error 0.10), with a first-stage F -statistic of 39.7 and an Anderson–Rubin 95% confidence set of $[0.44, 0.88]$, which contains the single-homing estimate of 0.55. The implied price sensitivity is $\hat{\alpha} = 0.0102$ (bootstrap standard error 0.0015) against 0.0099 in the main specification.

Table O.4: Market step estimates under the multi-homing specification

Panel A: Parameter estimates		Panel B: First stage	
	IV		$\log(s_{jt}^{\text{inc}})$
$\log(s_{jt})$	0.73	$\tilde{z}_{jt}$	1.69
	(0.10)		(0.27)
eharmony	-0.61	F	39.73
	(0.22)		
match	-0.27		
	(0.15)		
okcupid	-2.16		
	(0.43)		
pof	-1.20		
	(0.31)		
$p_j(\hat{\alpha})$	0.0102		
	(0.0015)		

Notes: the first-stage regression of Panel B is residualized on site indicators. Standard errors appear in parentheses; that of α is from a parametric bootstrap over the market-step coefficients, as in the main specification.

I repeat the merger and interoperability analysis under the multi-homing model. Table O.5 reports the results. Relative to the main specification with single-homing, it remains

the case that a full merger combining joint pricing and interoperability is the consumer-welfare-maximizing regime. The other displayed results qualitatively mirror those from the single-homing specification.

Table O.5: Joint pricing and interoperability under multi-homing

Joint pricing	Mean price		Δ consumer welfare	
	No interop.	Interop.	No interop.	Interop.
No	\$59.00	\$89.66	0	\$3.21
Yes	\$62.62	\$81.80	\$-2.13	\$8.66

Notes: welfare changes are in dollars per three-month subscription period, per dating website user in the baseline equilibrium.

O.11 Micro step results

Table O.6 provides results of the micro step estimation for the baseline specification. Table O.8 provides results of the micro step estimation for the “Age” demographic group specification. For each specification, standard errors are computed from the inverse Hessian estimate of the estimator’s asymptotic variance.

O.12 Robustness: instruments using demographic variation

The instruments of the main specification use no demographic variation. This appendix reports a robustness specification in which the Waldfogel predictions are instead computed from the full set of consumer characteristics, including both demographics and internet measures, and in which the market-level mean of every characteristic entering the prediction is added to (16) as a control. Under this specification, the exclusion restriction is imposed conditional on the mean consumer characteristics. Conditional on these controls, the instrument varies only through the interactions of demographics and web usage variables as entered in the nonlinear logit function used to predict site usage.

This specification has a stronger first stage (F -statistic of 28.8) and yields a network externality estimate of $\hat{\eta} = 0.43$ (standard error 0.13), statistically indistinguishable from the main estimate. Restricting the instrument basis to the web usage variables while retaining the full control set yields $\hat{\eta} = 0.49$ (standard error 0.17). The two sets of instruments jointly pass the Sargan overidentification test (p -values of 0.44 in the “Overall” specification and 0.84 in the “Age” specification). Table O.10 reports the full estimates.

O.13 Inferring the price sensitivity parameter

This appendix provides details of the procedure, described in Section 4 of the main text, for estimating the price sensitivity parameter α from the equilibrium first-order condition for prices. The difficulty is that the first-order condition involves the derivative of equilibrium market shares with respect to prices, which itself depends on α . The key observation is

Table O.6: First-stage parameter estimates – “Overall” demographic group specification, demographic variables

	eharmony.com	match.com	okcupid.com	pof.com
Education: High school or less (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Education: Some college	0.072	0.106	-0.071	0.369
	(0.150)	(0.103)	(0.326)	(0.228)
Education: College degree	-0.076	-0.063	-0.293	-0.206
	(0.161)	(0.109)	(0.355)	(0.273)
Education: Advanced degree	-0.028	0.048	-1.329	-0.139
	(0.182)	(0.121)	(0.627)	(0.304)
Education: Unknown	0.230	0.020	-0.356	0.462
	(0.112)	(0.078)	(0.238)	(0.179)
Age: Under 25yo (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Age: 25-29yo	0.141	0.285	-0.110	0.344
	(0.178)	(0.135)	(0.411)	(0.253)
Age: 30-34yo	0.063	0.294	-0.460	0.424
	(0.167)	(0.126)	(0.402)	(0.238)
Age: 35-39yo	0.026	0.169	-0.539	0.266
	(0.163)	(0.124)	(0.382)	(0.236)
Age: 40-49yo	0.108	0.215	-0.216	0.013
	(0.154)	(0.118)	(0.343)	(0.226)
Age: 50+yo	0.097	0.249	-0.326	0.187
	(0.153)	(0.117)	(0.343)	(0.223)
Children in HH: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Children in HH: Yes	0.094	0.055	0.028	-0.045
	(0.067)	(0.050)	(0.180)	(0.097)
Race: White (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Race: Black	-0.082	-0.496	-0.587	-0.369
	(0.089)	(0.077)	(0.322)	(0.150)
Race: Asian	-0.287	-0.185	0.252	-0.993
	(0.216)	(0.142)	(0.388)	(0.510)
Race: Other	-0.293	-0.280	0.228	0.170
	(0.192)	(0.136)	(0.392)	(0.247)
Broadband: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Broadband: Yes	-0.496	-0.116	-0.131	-0.570
	(0.106)	(0.093)	(0.372)	(0.142)
Hispanic: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Hispanic: Yes	-0.107	0.069	-0.085	-0.208
	(0.059)	(0.043)	(0.168)	(0.091)
Income: Under 25k (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Income: 25-75k	0.089	-0.024	0.048	0.077
	(0.064)	(0.047)	(0.180)	(0.090)
Income: 75-100k	0.068	-0.054	0.061	-0.048
	(0.080)	(0.059)	(0.222)	(0.117)
Income: Over 100k	0.001	-0.052	0.306	-0.297
	(0.079)	(0.057)	(0.204)	(0.120)
HH size: 1 (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
HH size: 2	-0.136	-0.206	-0.228	-0.008
	(0.098)	(0.072)	(0.255)	(0.146)
HH size: 3	-0.323	-0.295	-0.318	-0.037
	(0.117)	(0.086)	(0.306)	(0.172)
HH size: Over 3	-0.233	-0.357	-0.438	-0.056
	(0.114)	(0.084)	(0.300)	(0.169)
Log local population	0.042	-0.023	-0.097	-0.061
	(0.021)	(0.015)	(0.054)	(0.029)

Table O.7: First-stage parameter estimates – “Overall” demographic group specification, web usage variables

	eharmony.com	match.com	okcupid.com	pof.com
Log Pages Viewed	0.141 (0.073)	0.237 (0.057)	0.548 (0.201)	0.100 (0.104)
Log Browsing Duration	-0.136 (0.060)	-0.094 (0.044)	0.016 (0.154)	-0.022 (0.085)
Pages Viewed: Adult	0.062 (0.012)	0.042 (0.011)	-0.022 (0.050)	0.078 (0.012)
Pages Viewed: Advert	-0.011 (0.011)	-0.001 (0.005)	-0.001 (0.013)	-0.003 (0.012)
Pages Viewed: Finance	0.030 (0.027)	0.017 (0.019)	-0.089 (0.084)	-0.126 (0.050)
Pages Viewed: Gaming	-0.002 (0.011)	-0.033 (0.011)	-0.035 (0.032)	-0.012 (0.020)
Pages Viewed: Government	0.018 (0.028)	0.010 (0.021)	-0.676 (0.346)	-0.108 (0.088)
Pages Viewed: Info	-0.061 (0.053)	0.021 (0.030)	0.140 (0.046)	-0.087 (0.087)
Pages Viewed: Malware	0.009 (0.007)	-0.017 (0.006)	-0.041 (0.023)	-0.013 (0.014)
Pages Viewed: Media	-0.054 (0.021)	0.008 (0.011)	0.043 (0.021)	-0.174 (0.044)
Pages Viewed: Other	0.000 (0.001)	-0.003 (0.001)	0.001 (0.001)	0.001 (0.001)
Pages Viewed: Portal	0.016 (0.004)	0.033 (0.003)	0.011 (0.009)	0.024 (0.005)
Pages Viewed: Retail	0.002 (0.004)	0.002 (0.003)	-0.000 (0.010)	0.004 (0.006)
Pages Viewed: Social Media	-0.004 (0.001)	-0.003 (0.001)	-0.004 (0.003)	0.002 (0.002)
Pages Viewed: Video	-0.021 (0.012)	-0.070 (0.011)	0.025 (0.013)	-0.076 (0.025)
Pages Viewed: Weather	-0.102 (0.074)	-0.053 (0.043)	-0.135 (0.179)	-0.193 (0.119)
Pages Viewed: Webservice	0.007 (0.008)	-0.004 (0.007)	0.010 (0.014)	-0.048 (0.023)
Pages Viewed: Internet/Wireless	-0.006 (0.025)	-0.001 (0.018)	-0.021 (0.065)	-0.043 (0.050)
Pages Viewed: News	-0.185 (0.076)	0.063 (0.031)	-0.026 (0.123)	0.047 (0.064)
Pages Viewed: Sports	-0.123 (0.049)	0.011 (0.022)	-0.123 (0.103)	-0.043 (0.056)
Pages Viewed: Travel	0.133 (0.079)	0.205 (0.055)	-0.992 (0.417)	-0.108 (0.159)
Pages Viewed: Career	0.205 (0.062)	0.102 (0.054)	-0.127 (0.275)	0.166 (0.095)
Pages Viewed: Downloads	0.110 (0.108)	-0.102 (0.109)	-0.388 (0.495)	0.160 (0.146)
Pages Viewed: Directory	0.898 (0.431)	0.739 (0.372)	1.033 (1.120)	1.084 (0.593)

Table O.8: First-stage parameter estimates (“Age” demographic group specification), demographic variables

	eharmony.com	match.com	okcupid.com	pof.com
Education: High school or less (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Education: Some college	0.071	0.112	-0.068	0.355
	(0.148)	(0.102)	(0.323)	(0.224)
Education: College degree	-0.078	-0.060	-0.295	-0.214
	(0.159)	(0.109)	(0.353)	(0.270)
Education: Advanced degree	-0.027	0.056	-1.313	-0.146
	(0.181)	(0.120)	(0.625)	(0.301)
Education: Unknown	0.228	0.024	-0.350	0.453
	(0.110)	(0.076)	(0.235)	(0.173)
Age: Under 25yo (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Age: 25-29yo	0.149	0.319	-0.094	0.357
	(0.171)	(0.131)	(0.407)	(0.244)
Age: 30-34yo	0.059	0.321	-0.451	0.428
	(0.161)	(0.122)	(0.397)	(0.228)
Age: 35-39yo	-2.006	-2.042	-7.552	-2.154
	(0.156)	(0.120)	(0.376)	(0.226)
Age: 40-49yo	-1.923	-1.995	-7.236	-2.409
	(0.146)	(0.113)	(0.335)	(0.214)
Age: 50+yo	-1.933	-1.962	-7.343	-2.231
	(0.145)	(0.112)	(0.334)	(0.210)
Children in HH: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Children in HH: Yes	0.093	0.056	0.017	-0.041
	(0.067)	(0.050)	(0.180)	(0.097)
Race: White (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Race: Black	-0.085	-0.497	-0.584	-0.369
	(0.087)	(0.076)	(0.318)	(0.148)
Race: Asian	-0.286	-0.187	0.281	-0.996
	(0.213)	(0.140)	(0.381)	(0.508)
Race: Other	-0.300	-0.282	0.248	0.172
	(0.189)	(0.135)	(0.383)	(0.242)
Broadband: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Broadband: Yes	-0.498	-0.115	-0.131	-0.572
	(0.106)	(0.093)	(0.368)	(0.142)
Hispanic: No (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Hispanic: Yes	-0.108	0.069	-0.084	-0.206
	(0.059)	(0.042)	(0.166)	(0.090)
Income: Under 25k (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
Income: 25-75k	0.081	-0.026	0.052	0.082
	(0.064)	(0.047)	(0.178)	(0.090)
Income: 75-100k	0.063	-0.057	0.066	-0.046
	(0.079)	(0.059)	(0.221)	(0.116)
Income: Over 100k	-0.004	-0.053	0.316	-0.294
	(0.078)	(0.056)	(0.201)	(0.118)
HH size: 1 (Omit.)	0.000	0.000	0.000	0.000
	-	-	-	-
HH size: 2	-0.142	-0.212	-0.228	-0.014
	(0.097)	(0.072)	(0.252)	(0.143)
HH size: 3	-0.328	-0.300	-0.306	-0.040
	(0.116)	(0.085)	(0.303)	(0.170)
HH size: Over 3	-0.239	-0.362	-0.439	-0.064
	(0.113)	(0.084)	(0.298)	(0.167)
Log local population	0.043	-0.023	-0.095	-0.062
	(0.016)	(0.012)	(0.044)	(0.023)

Table O.9: First-stage parameter estimates (“Age” demographic group specification), web usage variables

	eharmony.com	match.com	okcupid.com	pof.com
Log Pages Viewed	0.142 (0.067)	0.241 (0.051)	0.538 (0.174)	0.094 (0.096)
Log Browsing Duration	-0.136 (0.060)	-0.097 (0.044)	0.014 (0.152)	-0.020 (0.085)
Pages Viewed: Adult	0.063 (0.012)	0.042 (0.011)	-0.023 (0.049)	0.077 (0.012)
Pages Viewed: Advert	-0.011 (0.011)	-0.001 (0.006)	-0.001 (0.013)	-0.003 (0.012)
Pages Viewed: Finance	0.031 (0.026)	0.018 (0.019)	-0.093 (0.083)	-0.126 (0.050)
Pages Viewed: Gaming	-0.002 (0.011)	-0.033 (0.011)	-0.034 (0.032)	-0.010 (0.019)
Pages Viewed: Government	0.016 (0.028)	0.007 (0.021)	-0.742 (0.345)	-0.106 (0.086)
Pages Viewed: Info	-0.057 (0.053)	0.021 (0.030)	0.140 (0.046)	-0.090 (0.086)
Pages Viewed: Malware	0.009 (0.007)	-0.017 (0.006)	-0.042 (0.023)	-0.013 (0.014)
Pages Viewed: Media	-0.055 (0.021)	0.008 (0.011)	0.042 (0.020)	-0.174 (0.044)
Pages Viewed: Other	0.000 (0.001)	-0.004 (0.001)	0.001 (0.001)	0.001 (0.001)
Pages Viewed: Portal	0.016 (0.004)	0.033 (0.003)	0.012 (0.008)	0.023 (0.005)
Pages Viewed: Retail	0.002 (0.004)	0.002 (0.003)	0.000 (0.010)	0.005 (0.006)
Pages Viewed: Social Media	-0.004 (0.001)	-0.003 (0.001)	-0.004 (0.002)	0.002 (0.002)
Pages Viewed: Video	-0.022 (0.012)	-0.070 (0.011)	0.025 (0.012)	-0.076 (0.025)
Pages Viewed: Weather	-0.105 (0.074)	-0.057 (0.043)	-0.130 (0.179)	-0.193 (0.119)
Pages Viewed: Webservice	0.007 (0.008)	-0.004 (0.007)	0.011 (0.014)	-0.049 (0.023)
Pages Viewed: Internet/Wireless	-0.006 (0.025)	-0.002 (0.018)	-0.025 (0.066)	-0.042 (0.049)
Pages Viewed: News	-0.186 (0.076)	0.064 (0.031)	-0.037 (0.124)	0.052 (0.064)
Pages Viewed: Sports	-0.123 (0.049)	0.014 (0.022)	-0.122 (0.103)	-0.041 (0.056)
Pages Viewed: Travel	0.135 (0.079)	0.205 (0.055)	-0.998 (0.417)	-0.110 (0.158)
Pages Viewed: Career	0.204 (0.062)	0.101 (0.054)	-0.123 (0.272)	0.168 (0.094)
Pages Viewed: Downloads	0.106 (0.107)	-0.105 (0.109)	-0.387 (0.492)	0.151 (0.146)
Pages Viewed: Directory	0.878 (0.428)	0.714 (0.369)	0.940 (1.172)	1.091 (0.587)

Table O.10: Market step parameter estimates, instruments using demographic variation with composition controls

Panel A: Parameter estimates

	OLS	IV
$\log(s_{jt})$	0.93 (0.02)	0.43 (0.13)
eharmony	2.72 (3.18)	3.95 (7.14)
match	1.54 (3.18)	3.14 (7.14)
okcupid	-0.88 (3.18)	-0.66 (7.13)
pof	2.49 (3.18)	3.33 (7.13)
$p_j(\hat{\alpha})$		0.0121 (0.0025)

Panel B: First stage of IV regression

	$\widetilde{\log(s_{jt})}$
$\tilde{z}_{jt}$	1.76 (0.33)
F	28.79

Notes: estimates are from weighted two-stage least squares including market-composition controls (the market-level mean of every characteristic entering the instrument prediction); the first stage in Panel B is residualized on site indicators and the controls.

that, under the logit specification, this derivative is proportional to α so that the first-order condition can be solved for α in closed form.

First, define $\bar{\sigma}_{jt}^d(p, s)$ as the mean probability that a consumer of demographic group d in market t uses site j under prices p when they believes the prevailing market shares are s . The mean is taken over the distribution of consumer characteristics W_{it} and unobservables ε_{ijt} within group d in market t . The beliefs s need not coincide with the market shares consistent with the choice probabilities $\bar{\sigma}(p, s)$.

Let $\sigma_t(p)$ denote the equilibrium market shares under prices p , defined implicitly by the fixed-point condition

$$\bar{\sigma}_t(p, \sigma_t(p)) = \sigma_t(p). \quad (13)$$

Recall that I specify indirect utilities as

$$\delta_j^d = \bar{\psi}_j^d - \alpha P_j + f_j^d(S^o, S^1, \dots, S^D) + \Xi_j^d + \varepsilon_{ij},$$

so that network externalities may depend on both overall and demographic-specific market shares. This specification nests both shares-type and quantities-type models, as the functions f_j^d can implicitly depend on the populations of demographic groups within markets.

The first-order condition for prices involves the derivative of equilibrium market shares $\sigma_t(p)$ with respect to prices. The implicit function theorem applied to (13) implies that, provided $I - (\partial/\partial s)\bar{\sigma}_t(p, \sigma_t(p))$ is nonsingular,

$$\frac{\partial}{\partial p}\sigma_t(p) = \left[I - \frac{\partial}{\partial s}\bar{\sigma}_t(p, \sigma_t(p)) \right]^{-1} \frac{\partial}{\partial p}\bar{\sigma}_t(p, \sigma_t(p)), \quad (14)$$

where I is the $JD \times JD$ identity matrix, $(\partial/\partial p)\sigma_t$ is $JD \times J$, and $(\partial/\partial s)\bar{\sigma}_t$ is $JD \times JD$. The derivative $(\partial/\partial p)\sigma_t$ depends on α through (14). To solve for α , I exploit the fact that under the logit specification this dependence is proportional. When ε_{ijt} are iid type-1 extreme value, the partial derivatives of $\bar{\sigma}$ with respect to prices take the form

$$\begin{aligned} \frac{\partial \bar{\sigma}_j^d}{\partial p_j}(p, s) &= -\alpha \bar{\sigma}_j^d(p, s) \left(1 - \bar{\sigma}_j^d(p, s) \right), \\ \frac{\partial \bar{\sigma}_j^d}{\partial p_k}(p, s) &= \alpha \bar{\sigma}_j^d(p, s) \bar{\sigma}_k^d(p, s), \quad k \neq j. \end{aligned}$$

Every entry of $(\partial/\partial p)\bar{\sigma}_t$ is therefore proportional to α , so the matrix

$$\tilde{D}_p \bar{\sigma}_t(p, s) := \frac{1}{\alpha} \frac{\partial}{\partial p} \bar{\sigma}_t(p, s)$$

does not depend on α . Because $(\partial/\partial s)\bar{\sigma}_t$ does not depend on α either, it follows from (14) that

$$\frac{\partial}{\partial p}\sigma_t(p) = \alpha \tilde{D}_p \sigma_t(p), \quad \tilde{D}_p \sigma_t(p) := \left[I - \frac{\partial}{\partial s}\bar{\sigma}_t(p, \sigma_t(p)) \right]^{-1} \tilde{D}_p \bar{\sigma}_t(p, \sigma_t(p)),$$

where $\tilde{D}_p \sigma_t$ depends on market shares and the parameters of the choice model but not on α .

Some of the sites in the application are free to use; I impose the first-order conditions of the paid sites only. Let $\mathcal{J}^P = \{j \in \mathcal{J} : P_j > 0\}$ collect them, and write $J^P = |\mathcal{J}^P|$.

Paid site j 's first-order condition at equilibrium prices P^* is

$$\sum_t \sum_d M_t^d \sigma_{jt}^d(P^*) + \sum_t \sum_d M_t^d \frac{\partial \sigma_{jt}^d}{\partial p_j}(P^*) P_j^* = 0.$$

Substituting $(\partial/\partial p)\sigma_t = \alpha \tilde{D}_p \sigma_t$ and writing $\Delta_{jt}^d = (\tilde{D}_p \sigma_t(P^*))_{(j,d),j}$ for the relevant entry, the first-order condition becomes

$$\alpha = - \frac{\sum_t \sum_d M_t^d \sigma_{jt}^d(P^*)}{\sum_t \sum_d M_t^d \Delta_{jt}^d P_j^*}.$$

As this holds for each paid site, I recover α by averaging across them:

$$\hat{\alpha} = -\frac{1}{J^p} \sum_{j \in \mathcal{J}^p} \frac{\sum_t \sum_d M_t^d S_{jt}^d}{\sum_t \sum_d M_t^d \hat{\Delta}_{jt}^d P_j}, \quad (15)$$

where $\hat{\Delta}_{jt}^d$ is obtained by substituting observed market shares for $\bar{\sigma}_{jt}^d$ and first-stage estimates for the parameters of the choice model.

The computation of $\tilde{D}_p \sigma_t$ requires $(\partial/\partial s) \bar{\sigma}_t$, which enters through (14). Under the logit specification, the partial derivatives of $\bar{\sigma}_j^d$ with respect to market shares are, for demographic groups d and g and sites j and k :

$$\begin{aligned} \frac{\partial \bar{\sigma}_j^d}{\partial s_j^g}(p, s) &= \bar{\sigma}_j^d(1 - \bar{\sigma}_j^d) \left[\frac{\partial f_j^d}{\partial s_j} \frac{M_t^g}{M_t} + \frac{\partial f_j^d}{\partial s_j^g} \right] - \sum_{r \neq j} \bar{\sigma}_j^d \bar{\sigma}_r^d \left[\frac{\partial f_r^d}{\partial s_j} \frac{M_t^g}{M_t} + \frac{\partial f_r^d}{\partial s_j^g} \right], \\ \frac{\partial \bar{\sigma}_j^d}{\partial s_k^g}(p, s) &= \bar{\sigma}_j^d(1 - \bar{\sigma}_j^d) \left[\frac{\partial f_j^d}{\partial s_k} \frac{M_t^g}{M_t} + \frac{\partial f_j^d}{\partial s_k^g} \right] - \sum_{r \neq j} \bar{\sigma}_j^d \bar{\sigma}_r^d \left[\frac{\partial f_r^d}{\partial s_k} \frac{M_t^g}{M_t} + \frac{\partial f_r^d}{\partial s_k^g} \right], \end{aligned}$$

where $ds_j/ds_j^g = M_t^g/M_t$ follows from $s_j = \sum_d (M_t^d/M_t) s_j^d$. These derivatives, together with the price derivatives recorded above, are the inputs to the computation of $\hat{\Delta}_{jt}^d$ in (15).

O.14 Quantities model results

Table O.11: Market step parameter estimates – “Overall” demographic group specification, quantity-type model

Panel A: Parameter estimates

	OLS	IV
$\log(M_t s_{jt})$	1.00 (0.03)	0.54 (0.15)
$\log(M_t)$	-0.99 (0.03)	-0.55 (0.14)
eharmony	0.48 (0.12)	-0.29 (0.31)
match	-0.75 (0.12)	-1.17 (0.24)
okcupid	-2.99 (0.16)	-4.69 (0.57)
pof	0.30 (0.13)	-0.83 (0.41)
$p_j(\hat{\alpha})$		0.0101 (0.0027)

Panel B: First stage of IV regression

	$\widetilde{\log(M_t s_{jt})}$
$\tilde{z}_{jt}$	2.49 (0.60)
F	17.45

Notes: estimates are from weighted two-stage least squares, as in the main specification of Table 3; the first stage in Panel B is residualized on the site indicators.

O.15 Linear externality specification

This appendix describes estimation of a specification of the model in which market shares enter preferences linearly rather than through a log transformation, i.e., $f_j(S_t) = \eta S_{jt}$. Whereas the logarithmic function flattens as networks grow, the linear externality does not. This means that the two specifications extrapolate differently in counterfactuals that combine the merging sites' user bases. Estimating the market step with the same instruments as under the log specification yields $\hat{\eta} = 2.86$ (standard error 1.80) with a first-stage F -statistic of 18.5; the pricing first-order conditions then give $\hat{\alpha} = 0.0121$ (standard error 0.0114). Under interoperability, each merging site's externality term becomes $\eta(S_{1t} + S_{2t})$.

Table O.12 reports prices and consumer welfare under the four regimes of the main text's merger analysis. The welfare ranking of the regimes is identical to that in the baseline model; a full merger with both interoperability and joint pricing maximizes consumer welfare, interoperability under price competition benefits consumers, and joint pricing without interoperability raises prices by a similar amount as in the baseline model (about \$19, versus \$16) and harms consumers.

Table O.12: Joint pricing and interoperability under the linear externality specification

(a) Prices			(b) Consumer welfare (\$/baseline site user)		
Interoperability?			Interoperability?		
Joint pricing?	No	Yes	Joint pricing?	No	Yes
No	\$62.42	\$65.49	No	0	\$85.94
Yes	\$81.88	\$59.77	Yes	-\$17.80	\$110.33

Notes: the table provides the mean price of a three-month subscription period across **eharmony.com** and **match.com** as predicted by the linear-externality model under each regime.

Notes: the table provides the mean change in consumer welfare per dating website user in the baseline equilibrium with neither joint pricing nor interoperability, expressed in dollars per three-month subscription period.

O.16 Additional results

This appendix reports two variants of the market-share variance decomposition of Section 4, which in the main text removes each force one at a time relative to the baseline. Here I report a sequential decomposition, in which the three forces are removed cumulatively, and an order-free (Shapley) attribution that apportions the interactions among them.

The one-at-a-time decomposition in the main text isolates each force's effect relative to the baseline but does not characterize the interactions among them. The sequential decomposition above removes forces in a fixed order, so the contribution attributed to each force depends on its position in the sequence. To address these shortcomings, I recompute equilibrium market shares under each possible combination of removed forces and average each force's marginal contribution over the six possible removal orders in a Shapley decomposition. Table O.14 reports the results. Network externalities account for 0.48 of baseline

Table O.13: Sequential decomposition of geographic variation in market shares

Site	Standard deviation in market shares	Share of baseline standard deviation in market shares			
	Baseline	Baseline	No network externalities	No Ξ_{jt}	Equalize de- mographics
match.com	0.056	1.00	0.48	0.18	0.00
eharmony.com	0.042	1.00	0.47	0.28	0.00
pof.com	0.041	1.00	0.69	0.32	0.00
okcupid.com	0.014	1.00	1.32	0.77	0.00
Average	0.049	1.00	0.53	0.25	0.00

Notes: this table is based on market shares among the inside goods. For each site, the second column reports the standard deviation across markets of its baseline inside share. The remaining columns report the corresponding standard deviation under each sequential counterfactual. Under these counterfactuals, I successively remove network externalities, then also Ξ_{jt} , then also cross-market demographic differences. The “Average” row reports the weighted average of the site-level entries, where the weights are sites’ national baseline market shares.

dispersion at the weighted average (between 0.41 and 0.55 in every removal order). Thus, the finding that network externalities explain about half of cross-market dispersion is not an artifact of the removal order. The order-free attribution assigns unobserved tastes Ξ_{jt} a larger share than observed demographics (0.39 versus 0.13), and Ξ_{jt} ’s marginal contribution exceeds that of demographics under each removal order. The quantity model yields nearly identical shares (0.47, 0.40, and 0.13).

Table O.14: Order-free (Shapley) decomposition of geographic variation in market shares

Site	Shapley share of baseline dispersion		
	Network externalities	Ξ_{jt}	Demographics
match.com	0.52	0.37	0.11
eharmony.com	0.50	0.37	0.13
pof.com	0.41	0.40	0.19
okcupid.com	-0.02	0.76	0.26
Average	0.48	0.39	0.13

Notes: each entry reports the indicated force’s marginal contribution to reducing the cross-market standard deviation of the site’s inside share, averaged over the six orders in which the three forces can be removed and expressed as a share of the baseline standard deviation. The “Average” row reports the average of the site-level entries weighted by sites’ national baseline market shares.

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Table O.15: Sequential decomposition of geographic variation in market concentration

	Herfindahl–Hirschman index (/10,000)			
	Baseline	No network externalities	No Ξ_{jt}	Equalize demographics
Mean	4198	2989	2954	2961
SD	443	124	57	0

Notes: The table reports the mean and standard deviation across markets of the Herfindahl–Hirschman index (HHI) computed from market shares among the inside goods. Each column corresponds to the same sequential counterfactual environments as in Table O.13. The baseline column reports the HHI in the estimated model, and the remaining columns report the HHI after successively removing network externalities, unobserved mean tastes Ξ_{jt} , and cross-market demographic differences.

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